

WICHITA EISENHOWER, KS, USA

WMO: 724500

Lat: 37.648N

Lon: 97.430W

Elev: 1321

StdP: 14.01

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.9	12.4	-2.6	5.0	12.1	1.6	6.3	15.4	31.4	42.4	28.1	38.5	10.5	0	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	100.5	73.8	97.3	74.2	94.1	73.9	77.9	91.1	76.9	90.1	75.7	89.0	14.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.4	135.2	84.2	73.2	129.5	82.8	72.0	124.4	81.7	42.4	91.0	41.1	90.1	40.0	89.0	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.4	25.8	23.4	DB	1.2	104.6	4.7	3.1	-2.2	106.9	-5.0	108.7	-7.6	110.4	-11.1	112.7	
			WB	0.3	80.4	4.5	1.4	-3.0	81.4	-5.6	82.2	-8.2	83.0	-11.5	84.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.9	33.4	37.5	47.4	56.7	66.9	76.9	81.4	80.0	72.1	59.3	46.5	35.4
	DBStd	19.14	10.51	11.33	10.97	9.17	7.93	6.19	5.70	5.61	7.82	9.16	9.50	9.86
	HDD50	1763	519	371	177	39	1	0	0	0	1	24	172	459
	HDD65	4370	980	771	551	273	75	3	0	0	25	221	555	916
	CDD50	4653	4	20	96	241	526	808	975	930	664	313	69	7
	CDD65	1788	0	0	5	25	135	361	510	465	239	46	2	0
	CDH74	19499	0	2	50	271	1222	3867	6117	5185	2336	433	16	0
	CDH80	9375	0	0	7	59	418	1830	3273	2645	1015	126	2	0
Wind	WSAvg	11.3	11.0	11.5	12.7	13.5	11.7	11.8	10.3	9.7	10.5	11.1	11.3	10.7
Precipitation	PrecAvg	31.7	0.8	1.1	2.3	2.7	4.4	4.6	3.5	3.6	3.3	2.6	1.5	1.2
	PrecMax	53.8	3.2	3.9	9.2	10.0	13.1	10.5	9.7	12.0	13.0	9.4	5.9	4.7
	PrecMin	12.2	0.0	0.0	0.0	0.2	0.5	0.4	0.0	0.2	0.5	0.0	0.0	0.0
	PrecStd	7.6	0.7	0.9	1.8	1.9	2.7	2.3	2.3	2.6	2.9	2.0	1.3	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	72.4	79.6	85.9	93.5	100.9	105.2	104.1	99.0	89.1	76.1	65.6
		MCWB	51.7	54.7	59.9	65.2	70.5	73.8	73.6	72.7	71.6	67.7	59.9	54.2
	2%	DB	59.7	66.2	73.9	80.7	88.5	96.3	101.2	99.7	93.3	83.3	71.1	60.1
		MCWB	46.9	52.3	59.2	63.5	70.4	74.1	73.8	73.7	71.9	65.1	57.4	50.5
	5%	DB	54.5	60.7	69.6	76.8	84.6	93.0	97.9	96.2	89.9	78.9	66.0	55.2
		MCWB	44.7	49.6	56.9	61.9	69.2	73.8	74.7	74.0	71.5	64.0	55.0	47.3
	10%	DB	49.7	55.4	65.1	72.7	81.0	89.8	94.2	92.5	86.1	74.4	61.8	50.8
		MCWB	42.2	46.9	54.4	60.2	68.1	72.7	74.7	73.6	69.6	62.7	53.3	44.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	59.2	66.2	69.5	75.2	78.9	80.0	79.3	76.6	72.8	63.7	59.5
		MCDB	59.3	65.4	73.5	80.5	85.5	92.7	92.4	92.1	89.4	81.1	71.3	63.3
	2%	WB	49.6	55.1	62.5	67.3	73.3	77.2	78.0	77.6	74.6	70.0	60.1	52.9
		MCDB	56.6	62.4	69.3	75.7	83.5	90.3	91.2	90.7	87.2	77.8	66.4	56.4
	5%	WB	45.7	51.2	59.3	65.0	71.6	75.6	76.9	76.3	73.2	67.4	57.8	48.2
		MCDB	54.3	60.2	66.6	73.3	81.0	88.2	90.5	89.6	84.9	75.0	64.1	54.2
	10%	WB	42.2	46.4	55.5	62.3	69.8	74.3	75.8	75.1	71.8	64.2	54.6	44.2
		MCDB	49.5	55.2	63.8	70.7	78.6	86.1	89.6	88.1	82.9	72.0	61.1	50.5
Mean Daily Temperature Range	5% DB	MDBR	20.4	21.2	22.0	22.0	20.4	20.7	21.0	20.8	21.6	22.0	21.2	19.0
		MCDBR	29.2	29.9	29.0	27.4	24.6	24.4	26.0	26.0	25.1	27.3	27.1	26.4
		MCWBR	18.7	18.7	16.7	15.3	11.0	8.4	6.9	7.5	8.8	12.9	16.0	17.7
	5% WB	MCDBR	27.6	27.0	23.0	23.0	21.0	21.9	21.9	21.5	21.2	20.4	21.9	23.0
		MCWBR	18.9	18.6	16.0	14.9	11.2	9.1	8.0	8.3	8.7	12.3	15.5	17.5
Clear-Sky Solar Irradiance	taub	0.287	0.298	0.334	0.372	0.405	0.413	0.421	0.424	0.380	0.339	0.299	0.281	
	taud	2.499	2.479	2.403	2.322	2.306	2.313	2.329	2.276	2.377	2.479	2.520	2.524	
	Ebn at Noon	288	298	295	288	279	275	272	269	276	279	281	283	
	Edh at Noon	25	29	35	40	41	41	40	41	35	29	24	23	
All-Sky Solar Radiation	RadAvg	802	1032	1352	1696	1901	2169	2136	1914	1590	1178	870	686	
	RadStd	81	148	91	113	167	157	129	107	114	132	88	66	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+229	N/A	
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page